

B4 Nuclear and Particle Physics — Model Solutions, 2020

Question 1 — Nuclear form factor

Problem. Define the nuclear form factor $F(q)$, derive the spherical-symmetry expression and identify C . Compute $F(q)$ for (i) a thin spherical shell of radius R , (ii) a uniformly charged ball of radius R . Outline a scattering experiment to discriminate the two; comment on the student's surface-shell argument.

(a)(i) Definition

In Born-approximation elastic electron–nucleus scattering, the amplitude factorises into a Mott (point) amplitude times $F(\mathbf{q})$, the Fourier transform of the normalised charge density:

$$F(\mathbf{q}) = \frac{1}{Ze} \int \rho(\mathbf{r}) e^{i\mathbf{q}\cdot\mathbf{r}/\hbar} d^3r, \quad \frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{Mott}} |F(\mathbf{q})|^2. \quad (1)$$

Normalisation: $F(0) = 1$.

(a)(ii) Spherical symmetry

Take polar axis along $\mathbf{q}$ in (??); let $q \equiv |\mathbf{q}|/\hbar$:

$$F(q) = \frac{1}{Ze} \int_0^\infty r^2 \rho(r) dr \int_0^\pi e^{iqr \cos \theta} \sin \theta d\theta \int_0^{2\pi} d\phi.$$

Do the ϕ integral:

$$F(q) = \frac{2\pi}{Ze} \int_0^\infty r^2 \rho(r) dr \int_0^\pi e^{iqr \cos \theta} \sin \theta d\theta.$$

Standard angular integral:

$$\int_0^\pi e^{iqr \cos \theta} \sin \theta d\theta = \frac{2 \sin(qr)}{qr}.$$

Substitute:

$$F(q) = \frac{4\pi}{Ze q} \int_0^\infty r \rho(r) \sin(qr) dr, \quad C = \frac{4\pi}{Ze}.$$

(2)

(b)(i) Thin spherical shell

Charge on a shell at $r = R$:

$$\rho_{\text{sh}}(r) = \frac{Ze}{4\pi R^2} \delta(r - R).$$

Insert into (??):

$$F_{\text{sh}}(q) = \frac{4\pi}{Ze q} \int_0^\infty r \cdot \frac{Ze}{4\pi R^2} \delta(r - R) \sin(qr) dr.$$

Delta picks $r = R$:

$$F_{\text{sh}}(q) = \frac{4\pi}{Ze q} \cdot \frac{Ze}{4\pi R^2} \cdot R \sin(qR).$$

Cancel common factors:

$$\boxed{F_{\text{sh}}(q) = \frac{\sin(qR)}{qR}}. \quad (3)$$

Zeros at $qR = n\pi$; envelope $\propto (qR)^{-1}$.

(b)(ii) Uniform ball

Density $\rho_v(r) = 3Ze/(4\pi R^3)$ for $r \leq R$. Insert into (??):

$$F_v(q) = \frac{4\pi}{Ze q} \cdot \frac{3Ze}{4\pi R^3} \int_0^R r \sin(qr) dr.$$

Cancel $4\pi/(Ze)$:

$$F_v(q) = \frac{3}{qR^3} \int_0^R r \sin(qr) dr.$$

Integrate by parts ($u = r$, $dv = \sin(qr)dr$):

$$\int_0^R r \sin(qr) dr = \left[-\frac{r \cos(qr)}{q} \right]_0^R + \frac{1}{q} \int_0^R \cos(qr) dr.$$

Evaluate boundary and remaining integral:

$$\int_0^R r \sin(qr) dr = -\frac{R \cos(qR)}{q} + \frac{\sin(qR)}{q^2}.$$

Substitute:

$$F_v(q) = \frac{3}{qR^3} \cdot \frac{1}{q^2} [\sin(qR) - qR \cos(qR)].$$

Collect powers of qR :

$$\boxed{F_v(q) = \frac{3}{(qR)^3} [\sin(qR) - qR \cos(qR)]}. \quad (4)$$

First zero at $qR \approx 4.49$; envelope $\propto (qR)^{-2}$, secondary maxima rapidly suppressed.

(c) Discriminating experiment

Probe: high-energy electron beam (point-like, EM only, no strong-interaction complications). Need $\lambda \lesssim R$, i.e. $pc \gtrsim \hbar c/R \sim 40 \text{ MeV}$ for $R \sim 5 \text{ fm}$; choose $E_e \sim 300 \text{ MeV}$ so several diffraction minima are accessible. Ultra-relativistic kinematics:

$$q = \frac{2E_e}{\hbar c} \sin(\theta/2).$$

Geometry: thin nuclear target; magnetic spectrometer on movable arm selecting elastic electrons; sweep θ .

Observable: $d\sigma/d\Omega$ vs. θ ; divide by Mott cross-section to extract $|F(q)|^2$.

Discriminator (with R known): position of *first minimum* and decay rate of secondary maxima.

- Shell (??): first zero at $qR = \pi$ ($qR \approx 3.14$); maxima decay as $(qR)^{-2}$ in $|F|^2$.
- Volume (??): first zero at $qR \approx 4.49$; maxima decay as $(qR)^{-4}$, far steeper.

(d) Validity of the student's argument

Wrong. The strong nuclear force binds nucleons $\sim 100\times$ more strongly than Coulomb at fm scales, dwarfs proton–proton repulsion, and is charge-symmetric: protons sit in the same shell-model orbitals as neutrons. The proton density therefore tracks the (uniform) total nucleon density throughout the volume, with a Saxon–Woods edge of thickness ~ 0.5 fm. Hofstadter electron-scattering data confirm (??), not (??).

Question 2 — Beta decay and Cabibbo mixing

Problem. (a) State Fermi-theory assumptions and derive the Sargent Q^5 rule. (b) From the table of $J^P = \frac{1}{2}^+$ baryons ($n, \Lambda^0, \Sigma^0, \Lambda_c^+, \Lambda_b^0$): infer quark contents and explain Σ^0 's short lifetime. (c) Draw leading-order Feynman diagrams for the dominant $e^\pm$ -inclusive decay of Λ^0 and n . (d) Extract θ_C . (e) Discuss inter-generation mixing.

(a)(i) Fermi-theory assumptions

- Four-fermion contact interaction (point-like) at a single space-time vertex; coupling G_F .
- W -propagator local: $|q^2| \ll M_W^2$, so $1/(q^2 - M_W^2) \approx -1/M_W^2$ is constant.
- Allowed approximation: $|\mathcal{M}|^2$ independent of lepton momenta (zero relative orbital angular momentum).
- Massless neutrino, ultra-relativistic charged lepton.
- Coulomb correction to outgoing $e^\pm$ encoded in Fermi function $F(Z, E_e) \approx 1$.
- Recoiling daughter nucleus carries negligible kinetic energy ($M \gg E_e$).

(a)(ii) Sargent Q^5 rule

Fermi's Golden Rule:

$$d\Gamma = \frac{2\pi}{\hbar} |\mathcal{M}|^2 d\rho(E_f).$$

Three-body lepton phase space (recoil ignored):

$$d\rho \propto p_e^2 dp_e p_\nu^2 dp_\nu \delta(E_e + E_\nu - Q).$$

Massless neutrino: $E_\nu = p_\nu c = Q - T_e$:

$$p_\nu^2 = (Q - T_e)^2 / c^2.$$

Eliminate dp_ν via the delta ($dp_\nu = dE_\nu/c$):

$$\frac{d\Gamma}{dp_e} \propto p_e^2 (Q - T_e)^2.$$

Ultra-relativistic limit $T_e \rightarrow E_e = p_e c$:

$$\frac{d\Gamma}{dp_e} \propto p_e^2 (Q - p_e c)^2.$$

Integrate over allowed range $0 \leq p_e c \leq Q$:

$$\Gamma \propto \int_0^{Q/c} p_e^2 (Q - p_e c)^2 dp_e.$$

Rescale $x = p_e c / Q$, $dp_e = (Q/c) dx$:

$$\Gamma \propto \frac{Q^2}{c^2} \cdot Q^2 \cdot \frac{Q}{c} \int_0^1 x^2 (1 - x)^2 dx.$$

Collect powers of Q :

$$\Gamma \propto Q^5 \int_0^1 x^2 (1 - x)^2 dx.$$

The integral evaluates to $1/30$, a pure number:

$$\boxed{\Gamma \propto Q^5.} \quad (5)$$

(b)(i) Quark content

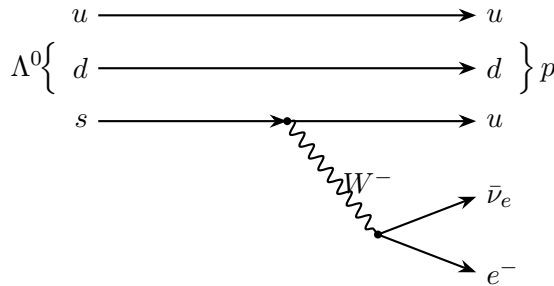
- $n = udd$ (standard nucleon, $I_3 = -\frac{1}{2}$).
- $\Lambda^0 = uds$ (isosinglet, antisymmetric flavour combination $(ud - du)s/\sqrt{2}$); long lifetime $\sim 10^{-10}$ s implies weak ($s \rightarrow u$) decay.
- $\Sigma^0 = uds$ (isotriplet $I_3 = 0$, symmetric combination $(ud + du)s/\sqrt{2}$); 77MeV above Λ^0 .
- $\Lambda_c^+ = udc$ (lightest charmed baryon: replace $s \rightarrow c$ in Λ^0).
- $\Lambda_b^0 = udb$ (lightest beauty baryon: replace $s \rightarrow b$ in Λ^0).

(b)(ii) Why Σ^0 is shortest-lived

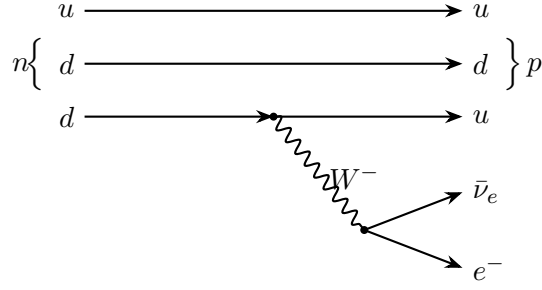
Σ^0 and Λ^0 share quark content. $\Sigma^0 \rightarrow \Lambda^0 \gamma$ is a flavour-conserving electromagnetic M1 transition: rate $\propto \alpha \sim 1/137$, no Cabibbo or phase-space suppression. The other four require flavour-changing weak decay ($\propto G_F^2$), giving lifetimes $\sim 10^{10}$ – 10^{14} longer.

(c) Feynman diagrams

(i) $\Lambda^0 \rightarrow p e^- \bar{\nu}_e$:



(ii) $n \rightarrow p e^- \bar{\nu}_e$: Identical topology with $s \rightarrow d$ on the bottom line.



Vertex couplings: $V_{us} = \sin \theta_C$ for Λ^0 , $V_{ud} = \cos \theta_C$ for n .

(d) Cabibbo angle

Assumptions:

1. Equal hadronic matrix elements (SU(3) flavour symmetry between $n \rightarrow p$ and $\Lambda \rightarrow p$).
2. Sargent Q^5 rule applies to both (allowed, no Coulomb correction).
3. Dominant e^- -inclusive mode is the simple semileptonic branch.

Partial widths:

$$\Gamma_n = K \cos^2 \theta_C Q_n^5, \quad \Gamma_\Lambda = K \sin^2 \theta_C Q_\Lambda^5.$$

Take ratio (eliminate K):

$$\tan^2 \theta_C = \frac{\Gamma_\Lambda}{\Gamma_n} \left(\frac{Q_n}{Q_\Lambda} \right)^5.$$

Compute partial widths from $\Gamma_i = \text{BR}_i \hbar / \tau_i$ with $\hbar = 6.582 \times 10^{-22} \text{ MeV s}$.

Neutron (BR= 1):

$$\Gamma_n = \frac{\hbar}{\tau_n} = \frac{6.582 \times 10^{-22}}{879.4} \text{ MeV}.$$

Evaluate:

$$\Gamma_n = 7.49 \times 10^{-25} \text{ MeV}.$$

Λ^0 (BR= 8.3×10^{-4}):

$$\Gamma_\Lambda = \frac{8.3 \times 10^{-4} \cdot 6.582 \times 10^{-22}}{2.63 \times 10^{-10}} \text{ MeV}.$$

Evaluate:

$$\Gamma_\Lambda = 2.08 \times 10^{-15} \text{ MeV}.$$

Q -values from $Q \simeq m_B - m_p - m_e$ ($m_p = 938.3$, $m_e = 0.51$, $m_n = 939.6$, $m_\Lambda = 1115.7 \text{ MeV}$):

$$Q_n = 939.6 - 938.3 - 0.51 = 0.78 \text{ MeV}.$$

$$Q_\Lambda = 1115.7 - 938.3 - 0.51 = 176.9 \text{ MeV}.$$

Compute Q -ratio:

$$\frac{Q_n}{Q_\Lambda} = \frac{0.78}{176.9} = 4.41 \times 10^{-3}.$$

Raise to fifth power:

$$(4.41 \times 10^{-3})^5 = 1.66 \times 10^{-12}.$$

Compute width ratio:

$$\frac{\Gamma_\Lambda}{\Gamma_n} = \frac{2.08 \times 10^{-15}}{7.49 \times 10^{-25}} = 2.78 \times 10^9.$$

Multiply:

$$\tan^2 \theta_C = 2.78 \times 10^9 \cdot 1.66 \times 10^{-12} = 4.61 \times 10^{-3}.$$

Take square root:

$$\tan \theta_C = \sqrt{4.61 \times 10^{-3}} = 0.0679.$$

Take arctangent:

$$\boxed{\theta_C \approx \arctan(0.068) \approx 3.9^\circ.} \quad (6)$$

Comment. Accepted value $\theta_C \approx 13^\circ$; the simple Q^5 rule under-estimates by ~ 3 because $Q_n \sim m_e$ violates the ultra-relativistic limit. Exact Fermi-integral f_n exceeds $Q_n^5/30$ by ~ 9 , restoring $\theta_C \approx 13^\circ$. The naive estimate (??) is the answer extractable from the table alone.

(e) Inter-generation mixing

The five baryons span all three generations. Reading lifetimes and electron-inclusive BRs:

- n : $|V_{ud}|^2 \approx \cos^2 \theta_C \approx 0.95$ (Cabibbo-favoured 1st-gen transition).
- Λ^0 : BR = 8.3×10^{-4} to e^- probes $|V_{us}|^2 \approx \sin^2 \theta_C \approx 0.05$. First-second generation mixing is small but nonzero.
- Σ^0 : EM decay, no CKM information.
- Λ_c^+ : dominant $c \rightarrow s$ Cabibbo-favoured, $|V_{cs}|^2 \approx \cos^2 \theta_C$. Semileptonic BR = 0.045 reflects ~ 1 in 3 light-quark final states.
- Λ_b^0 : dominant $b \rightarrow c$ with $|V_{cb}| \approx 0.04$ (third-second generation mixing, small but appreciable). $b \rightarrow u$ ($|V_{ub}| \approx 0.004$) further suppressed.

Wolfenstein hierarchy in $\lambda = \sin \theta_C \approx 0.22$:

$$|V_{ud}| \sim |V_{cs}| \sim 1, \quad |V_{us}| \sim |V_{cd}| \sim \lambda, \quad |V_{cb}| \sim \lambda^2, \quad |V_{ub}| \sim \lambda^3. \quad (7)$$

Mixing decreases sharply between non-adjacent generations.

Question 3 — γp cross section and the $\Delta(1232)$

Problem. The plot shows $\sigma_{\text{tot}}(\gamma p)$ vs. lab photon energy E_γ . (a) Lowest resonance: extract mass and lifetime. (b) Place its $J^P = \frac{3}{2}^+$ in the quark model. (c) Predict $\Gamma_{\gamma p}/\Gamma$ from the peak. (d) Interpret the flat plateau above 2 GeV. (e) Effect of switching to liquid deuterium.

(a) Mass and lifetime

Read peak at $E_\gamma \approx 0.32$ GeV. Mandelstam s for photon on stationary proton:

$$s = m_p^2 + 2m_p E_\gamma.$$

Compute $W = \sqrt{s}$ at peak (with $m_p = 0.938$ GeV):

$$W = \sqrt{(0.938)^2 + 2(0.938)(0.32)} \text{ GeV} = \sqrt{0.880 + 0.601} \text{ GeV}.$$

Evaluate:

$$W \approx 1.22 \text{ GeV}.$$

$$\boxed{M_\Delta \approx 1220 \text{ MeV}/c^2.} \quad (8)$$

Read FWHM in E_γ : peak $\sigma \approx 0.55 \text{ mb}$, half-max $\approx 0.27 \text{ mb}$ between $E_\gamma \sim 0.22$ and $\sim 0.42 \text{ GeV}$, so $\Gamma_{\text{lab}} \approx 0.20 \text{ GeV}$.

Differentiate $s = m_p^2 + 2m_p E_\gamma$:

$$2W \, dW = 2m_p \, dE_\gamma \implies \frac{dW}{dE_\gamma} = \frac{m_p}{W}.$$

Convert FWHM to width in W :

$$\Gamma = \frac{m_p}{W} \Gamma_{\text{lab}} = \frac{0.938}{1.22} \cdot 0.20 \text{ GeV}.$$

Evaluate:

$$\Gamma \approx 0.15 \text{ GeV} = 150 \text{ MeV}.$$

Lifetime from $\tau = \hbar/\Gamma$ with $\hbar = 6.582 \times 10^{-22} \text{ MeV s}$:

$$\tau = \frac{6.582 \times 10^{-22} \text{ MeV s}}{150 \text{ MeV}}.$$

Evaluate:

$$\tau \approx 4.4 \times 10^{-24} \text{ s}.$$

$$\boxed{\tau_\Delta \approx 4 \times 10^{-24} \text{ s}.} \quad (9)$$

Hadronic timescale: $\Delta \rightarrow N\pi$ via strong interaction.

(b) Quark model

The Δ couples to γp , so its quark content is uud (photon does not change flavour). $J^P = \frac{3}{2}^+$ requires aligned quark spins ($S = \frac{3}{2}$) and $L = 0$. Spatial wavefunction: symmetric. Spin wavefunction: symmetric. Colour wavefunction: antisymmetric (singlet ϵ_{abc}). Pauli requires the flavour wavefunction symmetric. Symmetric flavour uud with $I_3 = +\frac{1}{2}$ is the Δ^+ , member of the spin- $\frac{3}{2}$ baryon decuplet ($I = \frac{3}{2}$ multiplet $\Delta^{++}, \Delta^+, \Delta^0, \Delta^-$). Contrast with the proton: same uud but mixed-symmetry flavour-spin gives $J = \frac{1}{2}$, baryon octet.

(c) Branching ratio $\Gamma_{\gamma p}/\Gamma$

Breit–Wigner at $E = E_0$ gives factor $4\Gamma_i\Gamma_f/\Gamma^2$; sum over final states ($\sum_f \Gamma_f = \Gamma$):

$$\sigma_{\text{max}} = \frac{2J+1}{(2s_1+1)(2s_2+1)} \frac{\pi}{k^2} \frac{4\Gamma_i\Gamma}{\Gamma^2}.$$

Simplify:

$$\sigma_{\text{max}} = \frac{2J+1}{(2s_1+1)(2s_2+1)} \frac{4\pi}{k^2} \frac{\Gamma_i}{\Gamma}, \quad \Gamma_i = \Gamma_{\gamma p}.$$

Spin counting: $J = \frac{3}{2}$ gives $2J+1 = 4$; photon has 2 physical helicities, proton $2s_2+1 = 2$:

$$\frac{2J+1}{(2s_1+1)(2s_2+1)} = \frac{4}{2 \cdot 2} = 1.$$

γp CM momentum at the resonance:

$$k = \frac{W^2 - m_p^2}{2W} = \frac{(1.22)^2 - (0.938)^2}{2(1.22)} \text{ GeV}.$$

Evaluate:

$$k = \frac{0.609}{2.44} \text{ GeV} \approx 0.249 \text{ GeV}.$$

Convert to inverse length ($\hbar c = 0.197 \text{ GeV fm}$):

$$k = \frac{0.249}{0.197} \text{ fm}^{-1} \approx 1.265 \text{ fm}^{-1}.$$

Compute $4\pi/k^2$:

$$\frac{4\pi}{k^2} = \frac{4\pi}{(1.265)^2} \text{ fm}^2 \approx 7.85 \text{ fm}^2.$$

Convert ($1 \text{ fm}^2 = 10 \text{ mb}$):

$$\frac{4\pi}{k^2} \approx 78.5 \text{ mb}.$$

Insert into $\sigma_{\text{max}} = 1 \cdot (4\pi/k^2) \cdot B_{\gamma p}$:

$$\sigma_{\text{max}} \approx 78.5 \text{ mb} \cdot B_{\gamma p}.$$

Read $\sigma_{\text{max}} \approx 0.55 \text{ mb}$ (peak above zero); subtract non-resonant baseline $\sim 0.15 \text{ mb}$:

$$B_{\gamma p} \approx \frac{0.40}{78.5} \approx 5 \times 10^{-3}.$$

$$\boxed{\frac{\Gamma_{\gamma p}}{\Gamma} \approx 0.5\%}. \quad (10)$$

Consistent with PDG value $\sim 0.6\%$ for $\Delta \rightarrow N\gamma$.

(d) Flat plateau above 2 GeV

Reduced wavelength $\bar{\lambda} = \hbar c/E_\gamma \lesssim 0.1 \text{ fm}$ at $E_\gamma \gtrsim 2 \text{ GeV}$ resolves individual quarks/gluons; cross section becomes the incoherent sum of pointlike parton-level scatters. No further resonance peaks because the photon couples to free partons rather than exciting collective qqq states. The constant $\sigma \sim 0.13 \text{ mb}$ is the Bjorken-scaling/Pomeron-dominated total cross section, characteristic of incoherent point-like scattering.

(e) Switch to liquid deuterium

Deuteron contains $p + n$, weakly bound; cross section is incoherent sum:

$$\sigma(\gamma d) \approx \sigma(\gamma p) + \sigma(\gamma n).$$

By isospin, Δ photoexcitation is equally strong on p and n (both couple via the isovector M1 transition):

$$\sigma(\gamma n) \approx \sigma(\gamma p) \quad \text{at the } \Delta \text{ peak}.$$

At high energy, valence-quark counting also gives $\sigma(\gamma n) \approx \sigma(\gamma p)$ within $\sim 10\%$:

$$\boxed{\sigma(\gamma d) \approx 2\sigma(\gamma p)} \quad (11)$$

both on the resonance and on the plateau, with $\sim 5\text{--}10\%$ corrections from binding/shadowing. Resonance positions and plateau shape unchanged (isoscalar dynamics).

Question 4 — Semi-empirical mass formula

Problem. SEMF

$$M(A, Z) = Zm_p + (A - Z)m_n - \alpha A + \beta A^{2/3} + \gamma \frac{(A - 2Z)^2}{A} + \epsilon \frac{Z^2}{A^{1/3}} + \frac{\delta}{A^{1/2}},$$

with $\alpha = 15.8$, $\beta = 18.3$, $\gamma = 23.3$, $\epsilon = 0.714$ MeV, $\delta = \mp 12$ MeV (ee/oo) or 0 (eo/oe). (a) Estimate r_0 from ϵ . (b) Symmetric-fission energy: derive, evaluate for ^{235}U , account for the discrepancy with the observed 200 MeV. (c) Why thermal neutrons induce fission in ^{235}U but not ^{238}U . (d) Show $A \gtrsim 148$ implies α -instability. Given $m_\alpha = 3727.4$ MeV/ c^2 , electrostatic energy of uniform charged ball = $3Q^2/(20\pi\epsilon_0 R)$.

(a) Nuclear radius parameter r_0

Coulomb self-energy of uniform ball, charge Ze , radius $R = r_0 A^{1/3}$:

$$E_{\text{Coul}} = \frac{3}{5} \frac{(Ze)^2}{4\pi\epsilon_0 R}.$$

Substitute $R = r_0 A^{1/3}$:

$$E_{\text{Coul}} = \frac{3}{5} \frac{e^2}{4\pi\epsilon_0 r_0} \frac{Z^2}{A^{1/3}} \equiv \epsilon \frac{Z^2}{A^{1/3}}.$$

Read off coefficient ($\alpha_{\text{EM}} = e^2/(4\pi\epsilon_0 \hbar c)$):

$$\epsilon = \frac{3}{5} \frac{\alpha_{\text{EM}} \hbar c}{r_0}.$$

Solve for r_0 :

$$r_0 = \frac{3}{5} \frac{\alpha_{\text{EM}} \hbar c}{\epsilon} = \frac{3}{5} \cdot \frac{197.3 \text{ MeV fm}}{137.04 \cdot 0.714 \text{ MeV}}.$$

Evaluate denominator:

$$137.04 \cdot 0.714 = 97.85 \text{ MeV}.$$

Compute:

$$r_0 = \frac{3}{5} \cdot \frac{197.3}{97.85} \text{ fm} = \frac{3}{5} \cdot 2.016 \text{ fm}.$$

$$\boxed{r_0 \approx 1.21 \text{ fm.}}$$

(12)

(b)(i) Symmetric-fission energy

Parent $(A, Z) \rightarrow$ two daughters $(A/2, Z/2)$. Energy released:

$$Q_{\text{fis}} = M(A, Z) - 2M(A/2, Z/2).$$

Volume, asymmetry, and rest-mass terms are linear in A, Z and cancel exactly. Pairing term varies; ignore at leading order. Surface and Coulomb terms remain:

$$Q_{\text{fis}} = \beta [A^{2/3} - 2(A/2)^{2/3}] + \epsilon [Z^2/A^{1/3} - 2(Z/2)^2/(A/2)^{1/3}].$$

Reduce surface bracket: $2(A/2)^{2/3} = 2 \cdot 2^{-2/3} A^{2/3} = 2^{1/3} A^{2/3}$:

$$\beta [A^{2/3} - 2^{1/3} A^{2/3}] = -(2^{1/3} - 1) \beta A^{2/3}.$$

Reduce Coulomb bracket: $2(Z/2)^2/(A/2)^{1/3} = Z^2/2 \cdot 2^{1/3}/A^{1/3} = 2^{-2/3}Z^2/A^{1/3}$.

$$\epsilon[Z^2/A^{1/3} - 2^{-2/3}Z^2/A^{1/3}] = (1 - 2^{-2/3})\epsilon\frac{Z^2}{A^{1/3}}.$$

Combine:

$$Q_{\text{fis}} = (1 - 2^{-2/3})\epsilon\frac{Z^2}{A^{1/3}} - (2^{1/3} - 1)\beta A^{2/3}. \quad (13)$$

(b)(ii) $^{235}_{92}\text{U}$ evaluation

Compute powers ($A = 235$, $Z = 92$):

$$A^{1/3} = 6.171, \quad A^{2/3} = 38.16.$$

Compute Coulomb factor:

$$\frac{Z^2}{A^{1/3}} = \frac{8464}{6.171} = 1371.5.$$

Compute geometric coefficients:

$$1 - 2^{-2/3} = 1 - 0.6300 = 0.370, \quad 2^{1/3} - 1 = 0.260.$$

Coulomb gain:

$$0.370 \cdot 0.714 \cdot 1371.5 \text{ MeV} = 362.4 \text{ MeV}.$$

Surface cost:

$$0.260 \cdot 18.3 \cdot 38.16 \text{ MeV} = 181.5 \text{ MeV}.$$

Subtract:

$$Q_{\text{fis}}(^{235}\text{U}) \approx 362 - 182 = 181 \text{ MeV}. \quad (14)$$

(b)(iii) Discrepancy with 200 MeV

The SEMF estimate captures only the prompt kinetic-energy release at scission (Coulomb minus surface). The observed 200 MeV is the total energy from the full chain:

- Asymmetric split (mass-yield peaks at $A \approx 95, 140$, not $A = 117$): adds ~ 5 MeV via shell effects + asymmetry-term variation.
- Prompt neutron evaporation ($\sim 2\text{--}3$ neutrons, ~ 10 MeV total kinetic).
- Prompt γ -ray de-excitation of fragments (~ 8 MeV).
- Subsequent β^- decay of neutron-rich fragments to stability (~ 20 MeV, of which ~ 10 MeV in undetectable $\bar{\nu}_e$).

Sum of corrections ~ 20 MeV, raising calculated 181 MeV to observed 200 MeV.

(c) Thermal-neutron fission of ^{235}U vs. ^{238}U

Compound-nucleus excitation: $E^* = S_n + T_n$, where S_n is the neutron separation energy of the compound nucleus. Fission proceeds if $E^* > E_B \approx 6 \text{ MeV}$ (fission barrier).

$^{235}\text{U} + n \rightarrow ^{236}\text{U}^*$. Parent ^{235}U is even-odd ($Z = 92$ even, $N = 143$ odd, $\delta = 0$). Compound ^{236}U is even-even ($\delta = -12 \text{ MeV}$). Pairing energy released on absorption: extra $\sim 12 \text{ MeV}$ relative to even-odd. Result: $S_n(^{236}\text{U}) \approx 6.5 \text{ MeV} > E_B$. Thermal $T_n \sim 0.025 \text{ eV}$ already sufficient.

$^{238}\text{U} + n \rightarrow ^{239}\text{U}^*$. Parent ^{238}U even-even ($\delta = -12$). Compound ^{239}U even-odd ($\delta = 0$). Pairing now opposes absorption: S_n reduced by $\sim 12 \text{ MeV}$ compared to the even-even compound case. Result: $S_n(^{239}\text{U}) \approx 4.8 \text{ MeV} < E_B \approx 6.2 \text{ MeV}$. Need $T_n \gtrsim 1.4 \text{ MeV}$ to surmount barrier. The pairing term δ explains the dichotomy.

(d) α -instability for $A \gtrsim 148$

α -decay $(A, Z) \rightarrow (A - 4, Z - 2) + \alpha$ requires

$$Q_\alpha = M(A, Z) - M(A - 4, Z - 2) - m_\alpha > 0.$$

Nucleon rest-mass difference:

$$\Delta m_{\text{nuc}} = Zm_p + (A - Z)m_n - (Z - 2)m_p - (A - Z - 2)m_n = 2(m_p + m_n).$$

Subtract m_α :

$$2(m_p + m_n) - m_\alpha = 2(938.3 + 939.6) - 3727.4 \text{ MeV} = 28.3 \text{ MeV}.$$

This is the binding energy of the α particle, $B_\alpha \approx 28.3 \text{ MeV}$.

Define binding $B(A, Z) = \alpha A - \beta A^{2/3} - \gamma(A - 2Z)^2/A - \epsilon Z^2/A^{1/3}$ (drop pairing). Then:

$$Q_\alpha = B_\alpha + B(A - 4, Z - 2) - B(A, Z).$$

Taylor expand to first order in $\Delta A = 4$, $\Delta Z = 2$:

$$B(A - 4, Z - 2) - B(A, Z) \approx -4\partial_A B - 2\partial_Z B.$$

Compute $\partial_A B$ (∂_A of each term of B):

$$\partial_A B = \alpha - \frac{2\beta}{3A^{1/3}} + \gamma \frac{(A - 2Z)^2}{A^2} - 2\gamma \frac{A - 2Z}{A} + \frac{\epsilon Z^2}{3A^{4/3}}.$$

Compute $\partial_Z B$:

$$\partial_Z B = 4\gamma \frac{A - 2Z}{A} - \frac{2\epsilon Z}{A^{1/3}}.$$

Form $-4\partial_A B$:

$$-4\partial_A B = -4\alpha + \frac{8\beta}{3A^{1/3}} - \frac{4\gamma(A - 2Z)^2}{A^2} + \frac{8\gamma(A - 2Z)}{A} - \frac{4\epsilon Z^2}{3A^{4/3}}.$$

Form $-2\partial_Z B$:

$$-2\partial_Z B = -\frac{8\gamma(A - 2Z)}{A} + \frac{4\epsilon Z}{A^{1/3}}.$$

Add; the $\pm 8\gamma(A - 2Z)/A$ terms cancel:

$$-4\partial_A B - 2\partial_Z B = -4\alpha + \frac{8\beta}{3A^{1/3}} - \frac{4\gamma(A - 2Z)^2}{A^2} - \frac{4\epsilon Z^2}{3A^{4/3}} + \frac{4\epsilon Z}{A^{1/3}}.$$

Combine with B_α :

$$Q_\alpha = B_\alpha - 4\alpha + \frac{8\beta}{3}A^{-1/3} + \frac{4\epsilon Z}{A^{1/3}} - \frac{4\epsilon Z^2}{3A^{4/3}} - \frac{4\gamma(A-2Z)^2}{A^2}. \quad (15)$$

Term-by-term roles. Constant gain $B_\alpha - 4\alpha = 28.3 - 63.2 = -34.9$ MeV: net cost of removing four nucleons (volume term wins over α binding). Surface gain $\propto A^{-1/3}$ (decreasing). Coulomb gain $\propto Z/A^{1/3} \sim A^{2/3}$ (growing): this is the engine of instability. Asymmetry penalty mild.

Threshold along β -stability. Minimise SEMF in Z at fixed A :

$$0 = \frac{\partial M}{\partial Z} = (m_p - m_n) - 4\gamma \frac{A-2Z}{A} + \frac{2\epsilon Z}{A^{1/3}}.$$

Neglect $m_p - m_n \ll \gamma$, rearrange:

$$4\gamma \frac{A-2Z}{A} = \frac{2\epsilon Z}{A^{1/3}}.$$

Multiply by A and isolate Z :

$$4\gamma A = Z(8\gamma + 2\epsilon A^{2/3}).$$

Divide:

$$Z(A) \approx \frac{A/2}{1 + \epsilon A^{2/3}/(4\gamma)}.$$

Tabulate Q_α from (??):

A	120	140	148	160	200	235
Z	51	58	61	65	79	91
Q_α (MeV)	-1.8	-0.8	-0.2	+0.4	+2.8	+4.9

$Q_\alpha(A) \text{ changes sign near } A \approx 148.$

(16)

Sanity check. At $A = 148$, $Z = 61$, $A-2Z = 26$: contributions $-34.9 + 9.2 - 2.9 + 32.9 - 4.5 \approx -0.2$ MeV. Sign change at $A \sim 148$ confirmed.

Physical mechanism. Coulomb self-energy $\propto Z^2/A^{1/3} \sim A^{5/3}$ along β -stability; on emitting an α the fractional change $\sim 4/A$ liberates Coulomb energy growing as $A^{2/3}$. Combined with the unusually large α binding ($B_\alpha \approx 28$ MeV, doubly magic), it overcomes the volume cost of four nucleons at $A \sim 148$.

Caveat. Energetically allowed does not mean observable: the Coulomb barrier still must be tunnelled (Geiger–Nuttall law). At threshold ($A \sim 148$), half-lives are astronomical; observed α -emitters have $A \gtrsim 200$.